

Brendan Birozy

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EDUCATION

California State Polytechnic University, Pomona

Expected May 2027

B.S. Electrical Engineering - In Progress

EXPERIENCE

Formula SAE Electric “Bronco Electric Motorsports”, Cal Poly Pomona

Aug 2025 - Present

President, Chief Engineer

May 2026 - Present

- Lead overall strategy, execution, and vision for 50-member Formula SAE Electric team; coordinate people, priorities, resources, timelines, budget/schedule decisions, and compliance.
- Guide overall technical direction across all vehicle architectures while representing team in professional and campus engineering communities.

Electrical Design Engineer

Aug 2025 - Present

- Design, manufacture, test, validate, and iterate HV 400V and LV 12V/5V/3.3V boards for Battery Management System (BMS), including shutdown, precharge/discharge, indicator light, sensor, and embedded CAN/SPI/UART circuitry with Teensy4.1.
- Integrate PCBs into 96s4p tractive battery (six 16s4p Molicel P50B modules) and tractive system with Cascadia PM100DX inverter and Emrax 228MV motor as part of 5-member powertrain subteam and 30-member engineering team.
- Develop and build high- and low-voltage wiring harnesses across the vehicle.

System Lead, Business and Marketing

Sep 2025 - May 2026

- Lead sponsorship strategy, identify and secure industry/alumni partnerships, and draft business proposal and cost analysis submissions with engineering leads to maximize cost-to-performance ROI across subteams.
- Deliver spoken pitch decks to competition judges and help team place top 50 in business event; manage Framer website, sponsor communications, campus engagement, and LinkedIn/Instagram brand assets using Adobe, Canva, and Blender.
- Grow Instagram following from 250 (Oct 2025) to 1,200 (July 2026) while supporting recruitment, retention, and team photo/content production.

Parker Aerospace, Irvine, CA

May 2026 – August 2026

Electrical Engineering Intern, Commercial Flight Controls Division

- Convert sine-wave inverter driver board hardware designs from legacy FPGA-based architecture in DxDesigner to dissimilar FPGA and microcontroller-based control hardware in Altium.
- Test, analyze, and optimize signal integrity for high-frequency RS-485 and CAN transmission-line cable runs with isolated transceivers.
- Prototype IC implementations on legacy flight-control hardware by modifying existing PCBs with hand-built daughterboard circuitry; validate changes with FCC test consoles and test equipment incl. Oscilloscopes, Differential Probes, LCR meters, etc.
- Migrate DxDesigner schematics, symbols, and footprints into Altium libraries.

Autonomous Vehicle Laboratory, Cal Poly Pomona

Aug 2025 - Mar 2026

Embedded Systems Team Member

- Design and migrate drive-by-wire steering, throttle, and braking control systems prototypes to permanent PCBs.
- Work with 20-person team to integrate steering, throttle, and braking actuators and motors over CAN using RP2040s and Teensy4.1s; research 4WD hub motor powertrain strategies for off-road autonomous EV applications.

SKILLS

- Hardware: PCB Design, Schematic Capture, Circuit Analysis, Microcontrollers, Microelectronics, Power Electronics, High Voltage/High Current Design, Analog Design, Digital Design, Mixed Signal Design, ADC, FPGA, Digital Logic, Embedded Systems, Soldering, Lab Equipment, Fault/Failure Analysis, Signal Integrity, Power Electronics,
- Tools/Software: Altium, KiCAD, OrCAD, PSpice, LTspice, Digital Interfaces, MATLAB, Simulink, NI LabVIEW, Verilog, C/C++, CAN, SPI, UART, RS-485, SolidWorks, 3DEXperience, Microsoft SharePoint, Microsoft Office, Google Suite, Adobe Suite, Oscilloscope, Multimeter, Differential Probe, Multimeter, Logic Analyzer, Communication Bus Analyzer
- Battery Management Systems, Firmware, Digital Marketing, Project Management, Technical Communication, Technical Leadership, Cost Analysis, Team Leadership, Team Direction,